PROJECT PLAN

Project: Virtual Worker Platform

Technologies: n8n + Hugging Face

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1. PROJECT APPROACH

The Virtual Worker will be delivered in structured phases to ensure stability, governance, scalability, and enterprise readiness.

Core Principle: AI determines intent. Backend enforces policy. n8n executes controlled workflows.

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2. PHASE 0 – INFRASTRUCTURE FOUNDATION (Weeks 1–2)

Objectives

Establish secure and scalable technical environment.

Key Activities

- Setup Docker-based architecture
- Deploy n8n (queue mode enabled)
- Configure PostgreSQL database
- Implement CI/CD pipeline
- Setup centralized logging
- Configure Hugging Face API access
- Establish secret management process

Deliverables

- Running development and staging environments
- Secure credential vault
- Backup and recovery setup
- Monitoring baseline

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3. PHASE 1 – CORE TASK ENGINE (Weeks 3–6)

Objectives

Enable complete task lifecycle from intake to execution.

Key Activities

3.1 Task Management

- Build task submission UI
- Implement task state machine
- Create task database schema

3.2 AI Integration

- Integrate Hugging Face classification
- Implement entity extraction
- Implement confidence threshold logic

3.3 Template Engine

- Create template management system
- Map templates to n8n workflows
- Implement version control for templates

3.4 Execution Bridge

- Build backend-to-n8n webhook integration
- Capture workflow execution IDs
- Store execution results

3.5 Approval System

- Implement approval logic
- Create approval dashboard
- Enable resume-on-approval webhook

Deliverables

- 3–5 working automation templates
- End-to-end task execution
- Approval gating operational
- Audit logging enabled

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4. PHASE 2 – GOVERNANCE & HARDENING (Weeks 7–10)

Objectives

Prepare system for enterprise usage.

Key Activities

- Implement full RBAC model
- Enforce strict multi-tenant isolation
- Add observability dashboards
- Implement alerting for workflow failures
- Improve retry and timeout logic
- Conduct security validation testing

Deliverables

- Production-ready governance
- Security compliance checklist
- Monitoring and alerting dashboards

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5. PHASE 3 – SCALABILITY & OPTIMIZATION (Weeks 11–14)

Objectives

Ensure platform can scale and optimize costs.

Key Activities

5.1 n8n Scaling

- Enable horizontal scaling of workers
- Implement Redis queue management
- Load testing workflows

5.2 AI Cost Optimization

- Cache frequent AI outputs
- Reduce prompt sizes

- Evaluate self-hosted inference if cost threshold reached

5.3 Database Optimization

- Add indexing strategies
- Archive historical executions
- Optimize audit log storage

Deliverables

- Load testing report
- Performance benchmarks
- Cost monitoring dashboard

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6. RISK MANAGEMENT

Risk	Mitigation
AI misclassification	Confidence threshold + confirmation
Incorrect execution	Mandatory approvals
Data leakage	Workspace isolation enforcement
Workflow failure	Retry + alerting
Credential compromise	Vault + key rotation
Scaling bottlenecks	Queue mode + load testing

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7. KPI & SUCCESS METRICS

Technical KPIs

- 95% workflow success rate

- <5 second AI interpretation latency
- System uptime > 99%

Operational KPIs

- 50% reduction in manual task effort
- 70% of tasks executed without correction
- Average approval turnaround < 2 hours

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8. MVP EXIT CRITERIA

MVP considered complete when: - Users can submit tasks - AI correctly matches template - n8n executes workflow reliably - Approval flow pauses and resumes correctly - Results are returned to user - Full audit trail is visible

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9. LONG-TERM ROADMAP

Future enhancements may include: - Multi-agent coordination - Knowledge base integration - Advanced AI planning with constraint engine - Department-specific virtual workers - SaaS multi-organization scaling

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END OF PROJECT PLAN